TCET
DEPARTMENT OF VOCATIONAL STUDIES
Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education
Under Autonomy - CBCGS-HME 2025
University of Mumbai

NAME: IDRIS BOHRA **BATCH: 05**

SUBJECT: BUSINESS INTELLIGENCE – I **SYBVOC: SEMIII (SD)**

Report

On

Business Intelligence - I

By

Bohra Idris
S.Y B.VoC (Software Development)
Div: S.Y Batch E
Roll No: 01

TCET
THAKUR COLLEGE OF ENGINEERING & TECHNOLOGY

Choice Based Credit Grading System with Holistic Student Development

(CBCGS-H 2023)

Bachelor Of Vocation (2025-26)

Department Of Skill Development & Vocational Studies





CERTIFICATE

This is to certify that Mr. Idris Bohra Student of BVOC, Thakur College of Engineering and Technology, Mumbai. He have satisfactorily completed the requirements of Report on Business Intelligence as prescribe by **Thakur College of Engineering and Technology (An Autonomous College affiliated to University of Mumbai).**

Signature:_____.

Name:

Thakur College of Engineering and Technology, Kandivali (East) Mumbai.

Date:

Place:





INDEX PAGE

Sr. No	Titles	Page No.
01	Overview	4
02	Section 1- Introduction to Grocery Sales Data	5-7
03	Section 2- Data Collection and Preparation	8-11
04	Section 3- Data Modelling and DAX Measures	12-14
05	Section 4- Sales Dashboard and Business Analysis	15-17
06	Section 5- Advanced Power BI Analysis	18-22
07	Section 6- Expected Outcomes, Learning Outcomes and Skills Developed	23
08	Section 7- Conclusion	24

Overview: Business Intelligence (BI) is the process of collecting, cleaning, analyzing, and presenting data to support better business decisions. This project, **Grocery Sales Data Analysis using Microsoft Power BI**, focuses on transforming grocery sales data into meaningful information through interactive dashboards and visualizations.

The project uses sales transaction data along with related information about products, stores, employees, payment methods, and dates. Microsoft Power Query is used to prepare and organize the data, while the data model connects the different tables for analysis. DAX (Data Analysis Expressions) is used to create measures that calculate important business indicators such as total sales, total transactions, quantity sold, discounts, and return rate.

The Power BI report contains multiple dashboard pages covering overall sales performance, product analysis, store and employee performance, discounts and returns, customer and transaction analysis, and future sales forecasting. Charts, key performance indicators (KPIs), slicers, and advanced visuals help users explore the data from different perspectives.

The main objective of this project is to understand grocery sales patterns, identify high-performing products and stores, evaluate transaction trends, and present useful business insights through an interactive report. The project also provides practical experience in data cleaning, data modeling, DAX calculations, visualization, and forecasting using Microsoft Power BI.





Section 1 – Introduction to Grocery Sales Data

1.1 Introduction

The grocery retail industry generates large amounts of data through daily sales transactions. This data includes information about products, quantities sold, selling prices, discounts, payment methods, stores, employees, and transaction dates. Analyzing this information helps businesses understand sales performance and identify areas that require improvement.

Grocery Sales Data Analysis is the process of examining sales records to discover useful patterns and trends. It helps retailers identify popular product categories, compare store performance, evaluate employee contributions, and understand customer purchasing behaviour.

In this project, Microsoft Power BI is used to convert raw grocery sales data into an interactive Business Intelligence report. The data is prepared using Power Query, organized through relationships between tables, and analyzed using DAX measures and visualizations.

1.2 Importance of Grocery Sales Analysis

Grocery sales analysis is important for the following reasons:

- **Sales Monitoring:** Tracks total sales and transaction performance over time.
- **Product Performance:** Identifies high-performing products and categories.
- **Store Comparison:** Compares sales across different store locations.
- **Discount Analysis:** Examines discounts and their relationship with sales.
- **Employee Performance:** Evaluates sales contributions by employees.
- **Business Planning:** Supports decisions related to inventory, promotions, and future sales expectations.

By applying Business Intelligence techniques, grocery sales data can be transformed into meaningful information that supports effective business decisions.

1.3 Role of Microsoft Power BI

Microsoft Power BI is a business intelligence tool used to connect, clean, model, analyze, and visualize data. In this project, Power Query is used for data preparation, relationships are established between tables through data modelling, and DAX measures are created to calculate important business indicators.

The final report presents the analysis through interactive dashboards, charts, cards, slicers, tables, and advanced visuals. These features help users explore the data and understand business performance more effectively.

1.4 Dataset Description

The Grocery Sales Data project uses sales transaction records to analyze the performance of a grocery retail business. The dataset contains information about individual transactions and related details about products, stores, employees, payment methods, and dates.

The data is organized into related tables in Microsoft Power BI. This arrangement helps maintain a structured model and allows information from different tables to be analyzed together.

The major data fields used in the project include:

- **Transaction Details:** Transaction ID and transaction date.
- **Product Details:** Product ID, product name, and product category.
- **Store Details:** Store ID and store location.
- **Sales Details:** Quantity sold, unit price, discount percentage, and total sales.
- **Employee Details:** Employee ID, employee name, and contact information.
- **Payment Details:** Payment ID and payment mode.
- **Customer and Return Details:** Customer ratings and returned transaction status.
- **Date Details:** Year, quarter, month, day, and month-year information.

These fields support the calculation of key performance indicators such as total sales, total transactions, total quantity sold, average transaction value, total discount, and return rate.

1.5 Data Organization

The dataset is prepared and organized using Power Query before being used for reporting. Separate tables are maintained for sales, products, employees, stores, payments, and dates. Relationships between these tables allow the report to display results based on different business dimensions.

This structured approach improves the usability of the data and supports interactive filtering, comparison, and analysis across the dashboard pages.

1.6 Scope of the Project

The scope of the Grocery Sales Data Analysis project is to analyze grocery sales information using Microsoft Power BI and present the results through interactive dashboards. The project focuses on understanding historical sales performance and identifying patterns that can support business decisions.

The major areas covered in this project are:

- **Sales Analysis:** Examination of total sales, transaction counts, and quantity sold.
- **Product Analysis:** Comparison of sales and quantities across product categories and individual products.
- **Store and Employee Analysis:** Evaluation of sales performance by store and employee.
- **Discount and Return Analysis:** Study of discount amounts, discounted transactions, and returned transactions.
- **Customer and Transaction Analysis:** Examination of customer ratings and transaction values.
- **Sales Forecasting:** Use of Power BI forecasting capabilities to estimate future sales trends based on historical data.

1.7 Applications of the Project

The report can support several grocery retail business activities:

1. **Performance Monitoring:** Managers can monitor sales performance using key performance indicators.
2. **Product Planning:** Sales comparisons can help identify products and categories that perform well.
3. **Store Evaluation:** Store-level comparisons can help management identify differences in performance.
4. **Promotion Analysis:** Discount analysis can help evaluate promotional activity.
5. **Business Planning:** Historical trends and forecasts can assist with future planning.

1.8 Expected Benefits

The project provides a centralized and interactive view of grocery sales data. It reduces the need for manual analysis, simplifies comparisons, and presents business information in an understandable visual format. It also provides practical experience in data preparation, data modeling, DAX calculations, dashboard development, and forecasting using Microsoft Power BI.

The forecast results should be interpreted as estimates rather than guaranteed future outcomes.

Section 2 – Data Collection and Preparation

2.1 Introduction

Data collection and preparation are important stages in Business Intelligence because the quality of the analysis depends on the quality of the data. Raw data may contain missing values, duplicate records, inconsistent data types, and unnecessary columns that can affect the accuracy of the final report.

In this project, grocery sales data is prepared using Microsoft Power Query Editor in Power BI. The data is organized into relevant tables, and the required columns are selected, renamed, and transformed to make the dataset suitable for analysis.

2.2 Data Import

The grocery sales dataset is imported into Power BI to begin the analysis process. Power Query Editor is used to inspect the available tables and examine their structure, column names, data types, and data quality.

The project uses tables containing sales transactions and related information about products, employees, stores, payment methods, and dates. These tables provide the necessary information for creating a connected data model and developing interactive dashboards.

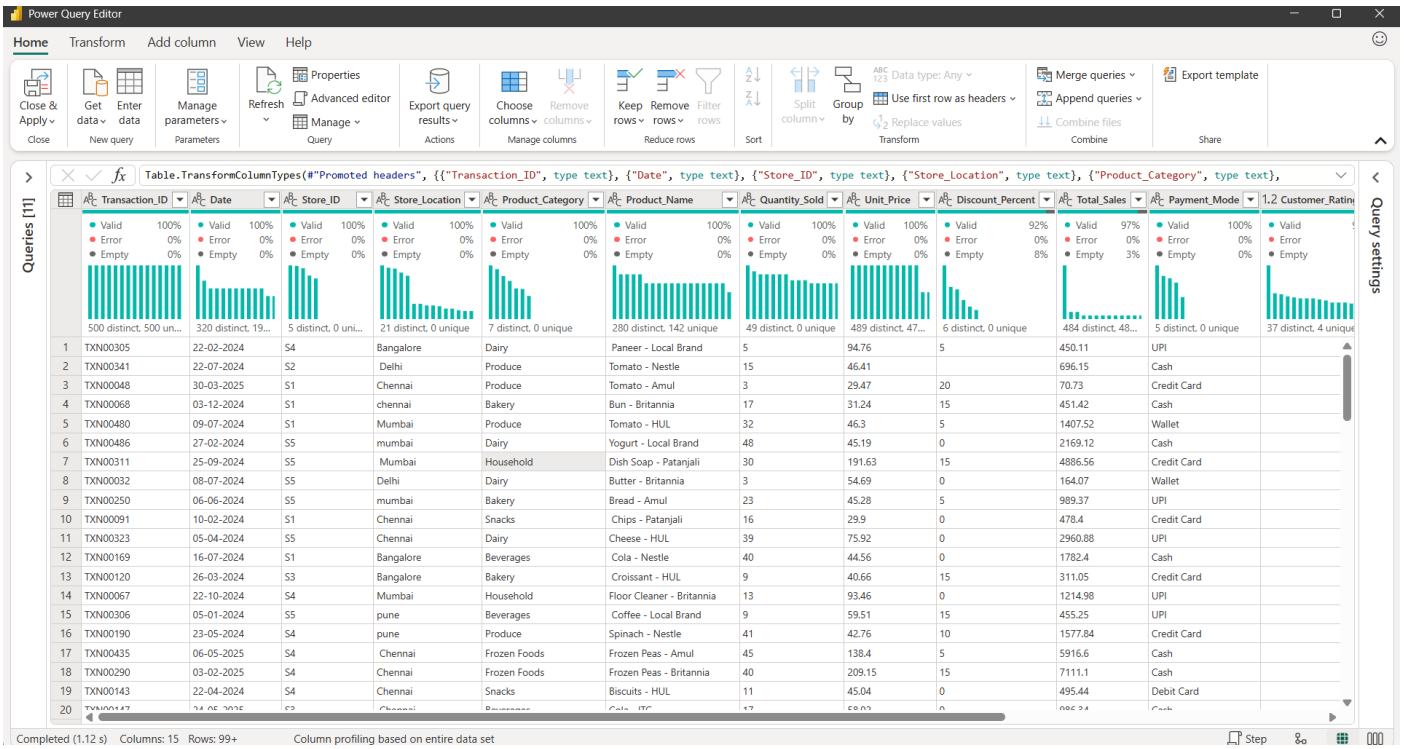


Figure 2.1: Grocery Sales Dataset in Power Query Editor

2.3 Need for Data Cleaning

Data cleaning helps improve consistency and reliability before analysis. The following checks are important during preparation:

- **Missing Values:** Identify blank or null values in important fields.
- **Duplicate Records:** Check for duplicate records and repeated entries.
- **Data Types:** Ensure dates, numeric values, and text fields use appropriate data types.
- **Column Names:** Use clear and consistent names for easier identification.
- **Data Quality:** Inspect errors, empty values, and inconsistent records.

These preparation steps help make the dataset more suitable for data modelling, DAX calculations, and visualization in Power BI.

2.4 Data Cleaning in Power Query

Microsoft Power Query Editor provides tools to clean and transform data before loading it into Power BI. In this project, the tables were inspected and prepared to improve consistency and support accurate analysis.

The main data preparation activities include:

- **Data Type Correction:** Columns were checked to ensure dates, numeric values, and text fields use appropriate data types.
- **Duplicate Checking:** Duplicate entries were examined to avoid unnecessary repetition in the data.
- **Column Preparation:** Relevant columns were retained and organized for analysis.
- **Missing Value Inspection:** Blank and null values were reviewed, particularly in important identification fields.
- **Date Preparation:** Date-related information was organized to support analysis by year, quarter, month, and date.
- **Table Organization:** Sales data and related information were prepared in separate tables for use in the data model.

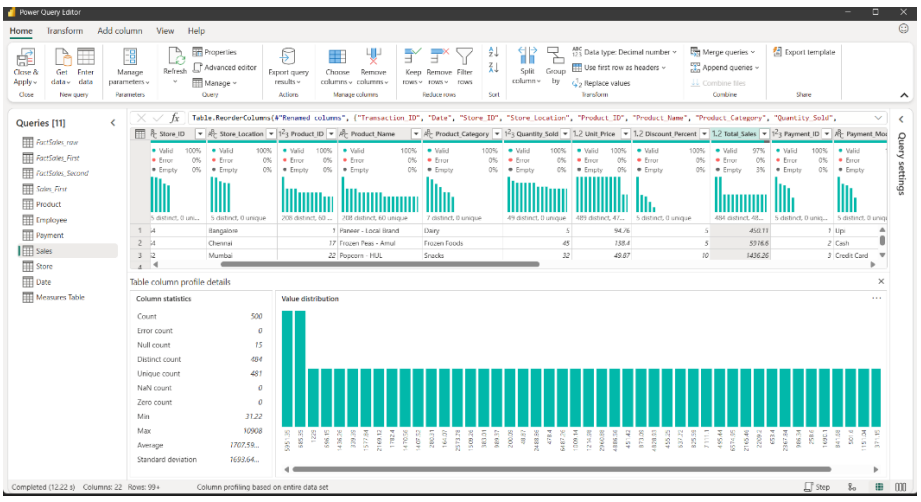


Figure 2.2: Data Cleaning and Preparation in Power BI

2.5 Handling Employee Information

Employee information requires careful preparation because employee names and identification numbers are used to analyze employee-level performance. Missing employee IDs and repeated employee names can affect the reliability of relationships and filtering.

During preparation, employee records were examined to identify missing or repeated information. The employee table was organized for use in the Power BI data model so that employee-related analysis could be performed more effectively.

2.6 Date Table Preparation

A date table supports time-based analysis by organizing calendar information into useful fields. The date table in this project includes fields such as date, year, month, and month-year. Sorting fields help display months in chronological order instead of alphabetical order.

The prepared date information supports sales trend analysis, year-to-date calculations, comparisons between periods, and the future sales forecasting visual.

2.7 Data Transformation

Data transformation converts the prepared dataset into a suitable format for analysis and visualization. In this project, Microsoft Power Query is used to organize the data before loading it into the Power BI data model. Properly structured data makes calculations and report creation easier.

The main transformation activities include:

- **Column Preparation:** Keeping relevant columns required for sales analysis.
- **Data Type Verification:** Ensuring dates, numerical values, and text fields use appropriate data types.
- **Date Formatting:** Organizing date-related information for monthly, quarterly, and yearly analysis.
- **Table Organization:** Keeping sales, product, store, employee, payment, and date information organized in suitable tables.
- **Data Consistency:** Reviewing column values and formats to support reliable analysis.

2.8 Data Loading

After preparation and transformation, the data is loaded into Microsoft Power BI for modeling and visualization. The loaded tables provide the foundation for creating relationships, calculating measures, and building interactive dashboards.

2.9 Importance of Data Preparation

Data preparation is an important stage of the Business Intelligence process because the quality of the final report depends on the quality and structure of the input data. Proper preparation helps reduce inconsistencies, supports accurate calculations, and makes the report easier to maintain and analyse.

In this project, data preparation establishes the foundation for the next stage: developing relationships between tables and creating meaningful sales measures.

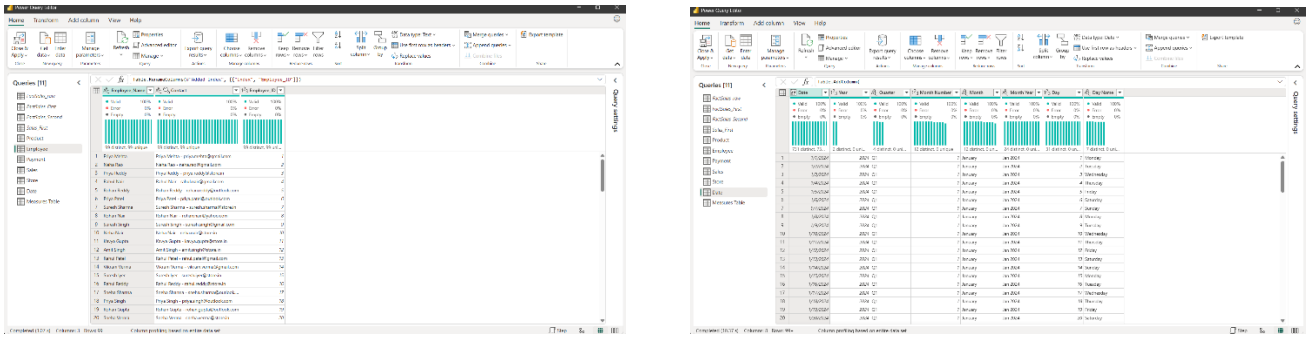


Figure 2.3: Prepared Grocery Sales Data in Power BI

Section 3: Data Modelling and DAX Measures

3.1 Introduction to Data Modelling

Data modelling is an important step in Business Intelligence that connects different tables and allows data to be analyzed efficiently. In this project, Microsoft Power BI is used to create a data model for the Grocery Sales Data Analysis report.

The dataset contains information about sales transactions, products, stores, employees, payment methods, and dates. Organizing these details into related tables helps avoid unnecessary repetition and supports meaningful analysis across different categories.

3.2 Need for Data Modelling

Data modelling helps establish connections between related tables so that users can analyze grocery sales from different perspectives. A properly organized model also supports accurate calculations and interactive filtering in Power BI.

The main purposes of data modelling are:

- To connect related tables using common fields or keys.
- To organize sales and supporting information systematically.
- To support filtering by product, store, employee, payment method, and date.
- To improve the structure of the report and its calculations.
- To provide a foundation for creating DAX measures and interactive dashboards.

3.3 Tables Used in the Data Model

The project uses sales transaction data along with related information about products, stores, employees, payment methods, and dates. These tables provide different details required for analyzing sales performance.

The relationships between tables allow information from different sources to be combined in Power BI visuals. The exact relationship fields and directions depend on the structure of the imported dataset.

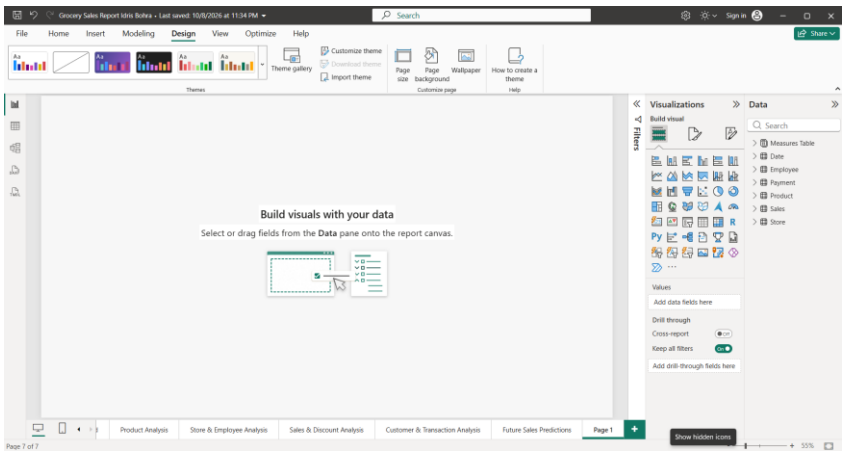


Figure 3.1: Grocery Sales Data Model in Power BI

3.4 Relationship Model

Data modeling establishes relationships between different tables so that information can be analyzed together. In this project, the relationship model is created in Microsoft Power BI to connect grocery sales transactions with the relevant product, store, employee, payment, and date information.

The Sales table acts as the central table in the model. The other tables provide additional information that helps analyze sales from different perspectives. A separate Measures Table is used to organize the DAX measures created for the report.

3.5 Tables in the Relationship Model

The main tables shown in the model are:

- **Sales Table:** Contains transaction-related information used for sales analysis.
- **Product Table:** Stores product details, including product ID, name, and category.
- **Store Table:** Contains store identification information.
- **Employee Table:** Contains employee IDs, names, and contact information.
- **Payment Table:** Stores payment IDs and payment modes.
- **Date Table:** Contains date-related fields such as day, month, quarter, and year.
- **Measures Table:** Organizes the measures used to calculate sales performance indicators.

3.6 Relationships and Their Importance

The relationship model connects the Sales table with the supporting tables through related fields. The model uses these connections to support filtering and analysis across products, stores, employees, payment methods, and dates.

The relationships help Power BI combine information from multiple tables while displaying results in report visuals. They also provide the foundation for time-based calculations, including Year-to-Date (YTD) Sales and Year-over-Year (YoY) Growth.

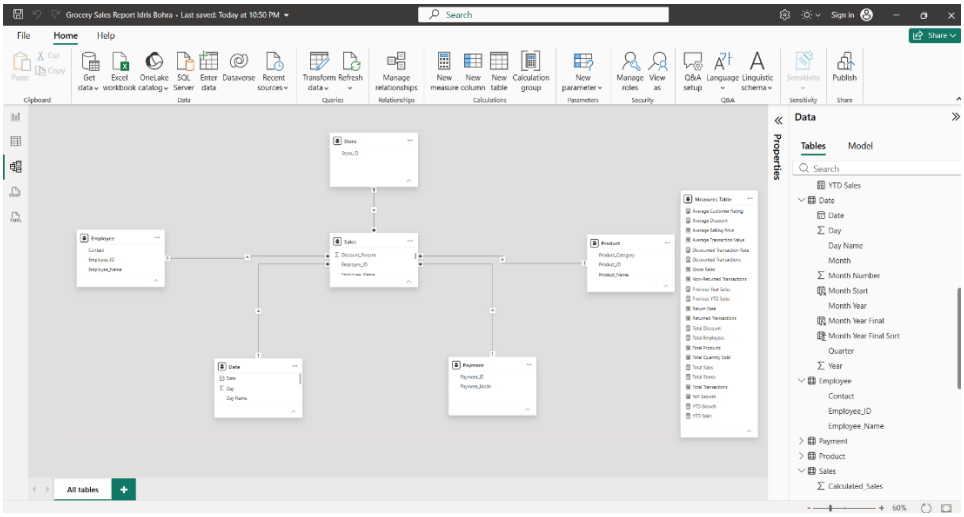


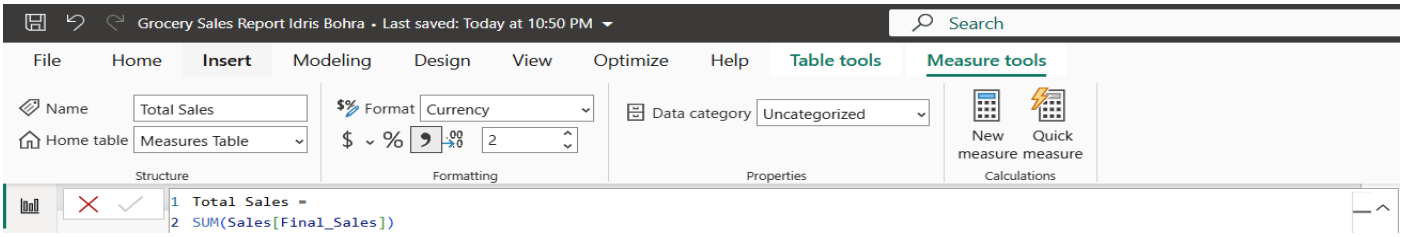
Figure 3.2: Relationship Model of the Grocery Sales Dataset in Power BI

The figure shows the tables and their relationships as configured in Microsoft Power BI Model view.

3.7 Implementation of DAX Measures

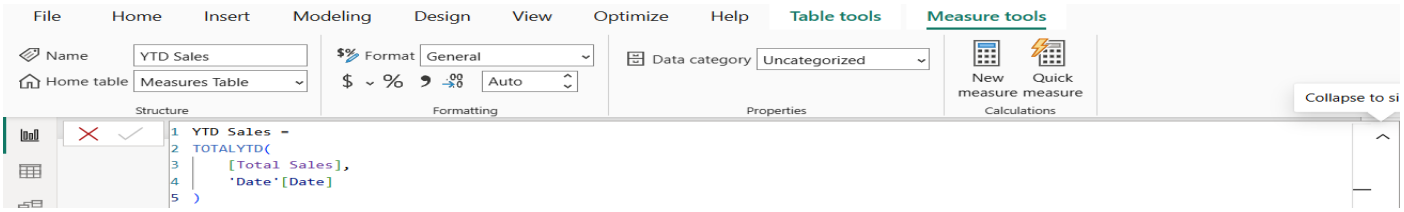
In this project, Data Analysis Expressions (DAX) is used to create measures that calculate sales performance and analyze changes over time. The following measures were implemented in Microsoft Power BI.

1. **Total Sales:**
   The Total Sales measure calculates the sum of the `Final_Sales` column from the Sales table.



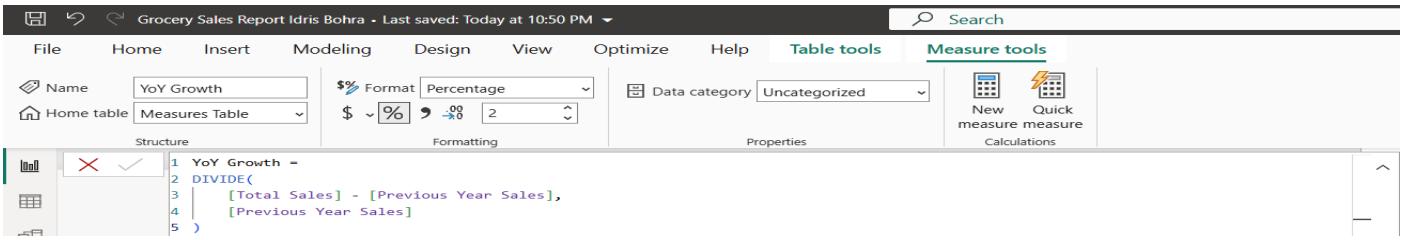
2. Year-to-Date (YTD) Sales:

The YTD Sales measure calculates the cumulative sales from the beginning of the year up to the selected date. It uses the Total Sales measure and the Date column from the Date table.



3. Year-over-Year (YoY) Growth:

The YoY Growth measure calculates the change in sales compared with the previous year's sales. The result is formatted as a percentage to make the growth rate easier to interpret.



3.8 Summary

DAX measures make the Grocery Sales Analysis report more useful by converting transaction data into meaningful performance indicators. Total Sales provides the overall sales figure, YTD Sales supports cumulative analysis, and YoY Growth helps evaluate changes between years.

Screenshots: Insert the three screenshots showing the actual DAX formulas for Total Sales, YTD Sales, and YoY Growth in the relevant places above.

Section 4: Sales Dashboard and Business Analysis

4.1 Introduction

The sales dashboard is an important component of the Grocery Sales Data Analysis project. It presents sales information through interactive visualizations created in Microsoft Power BI. The dashboard helps users understand overall sales performance, product contribution, transaction activity, and changes in sales over time.

The report uses Key Performance Indicators (KPIs), charts, tables, and slicers to make the information easier to understand. Users can explore the data by selecting different years, months, products, stores, and payment methods.

4.2 Objectives of Dashboard Analysis

The main objectives of this section are:

- To examine overall sales performance and transaction activity.
- To analyse sales trends over time.
- To compare sales across stores and product categories.
- To identify products contributing significantly to total sales.
- To explore sales information using interactive filters.
- To present useful business insights through visual representation.

4.3 Dashboard Overview

The following pages present the dashboards developed for the project. Each dashboard focuses on a particular aspect of grocery sales, allowing the data to be examined from different business perspectives. The analysis begins with the Executive Sales Dashboard and Product Analysis Dashboard, followed by more detailed analyses in Section 5.

4.4 Executive Sales Dashboard

The Executive Sales Dashboard provides an overview of the grocery business's sales performance. It displays important KPIs, sales trends, sales by store, sales by product category, and transactions by payment mode.

Key components of the dashboard:

- **Total Sales:** Displays overall sales of approximately $852.73K.
- **Total Transactions:** Shows 500 transactions.
- **Units Sold:** Displays approximately 13K units sold.
- **Average Transaction Value:** Shows approximately 1.71K.
- **Sales by Stores:** Compares sales performance across five stores.
- **Sales Trend Over Time:** Presents changes in sales from 2024 to 2025.
- **Sales by Product Category:** Compares the sales contribution of different categories.
- **Transactions by Payment Mode:** Shows the distribution of transactions across payment methods.

The dashboard also includes slicers for year, month, product name, store ID, and payment mode. These allow users to filter the report and investigate sales performance under different conditions.

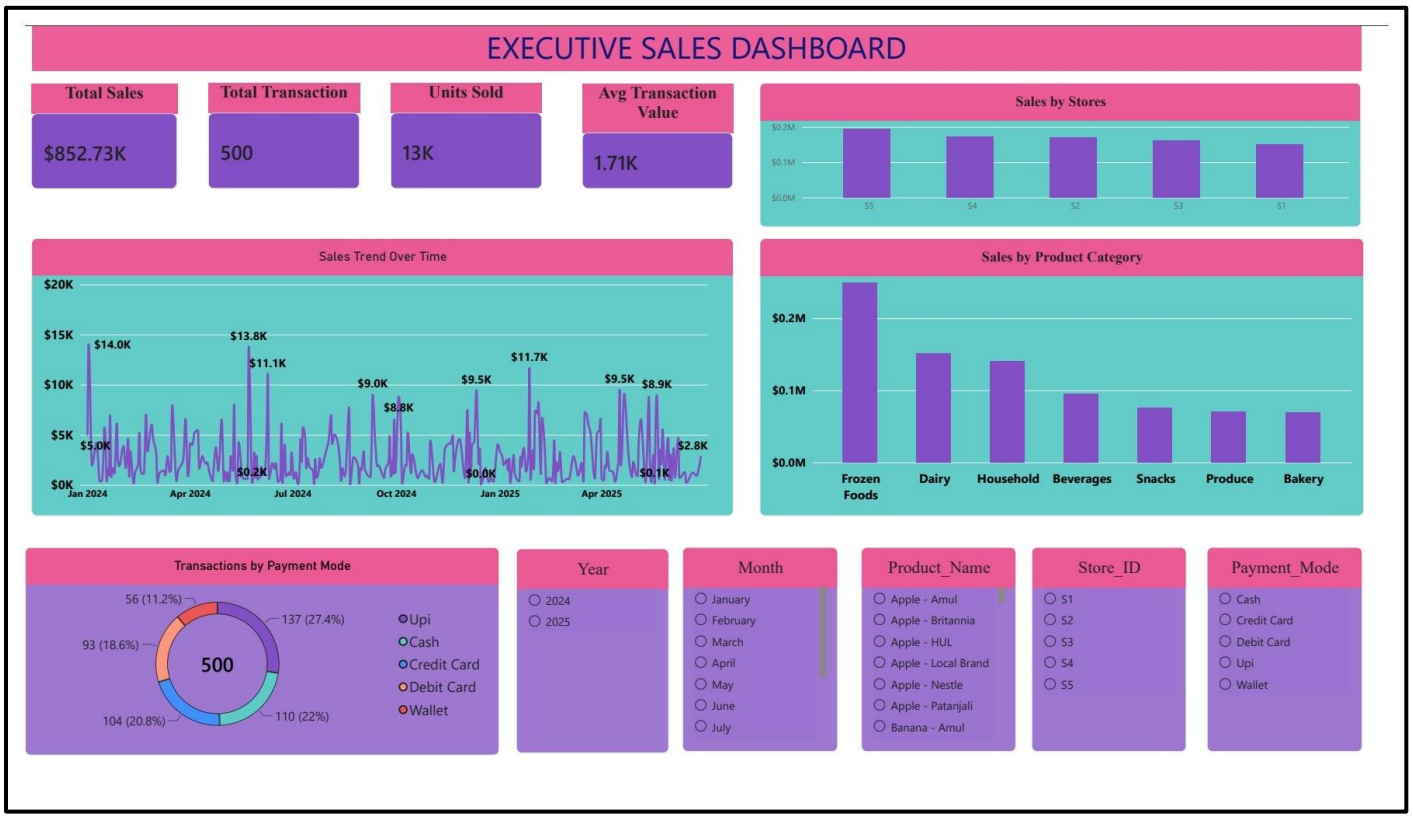


Figure 4.1: Executive Sales Dashboard in Power BI

4.5 Product Analysis Dashboard

The Product Analysis Dashboard focuses on product-level performance and comparisons between product categories. It helps identify which categories and products contribute most to sales and how much quantity is sold in each category.

Key components of the dashboard:

- **Total Sales:** Displays overall sales of approximately $852.73K.
- **Total Products:** Shows 208 products.
- **Units Sold:** Displays approximately 13K units sold.
- **Average Customer Rating:** Shows an average rating of 3.81.
- **Sales by Product Category:** Compares sales across categories such as Frozen Foods, Dairy, and Household.
- **Quantity Sold by Category:** Compares the quantity sold in each product category.
- **Category and Product Performance:** Presents sales, quantity sold, and average customer rating in a table.
- **Top Products by Sales:** Identifies products with comparatively high sales values.

A product-name slicer allows users to explore specific products. Together, these visuals help evaluate category performance, product demand, and customer ratings.

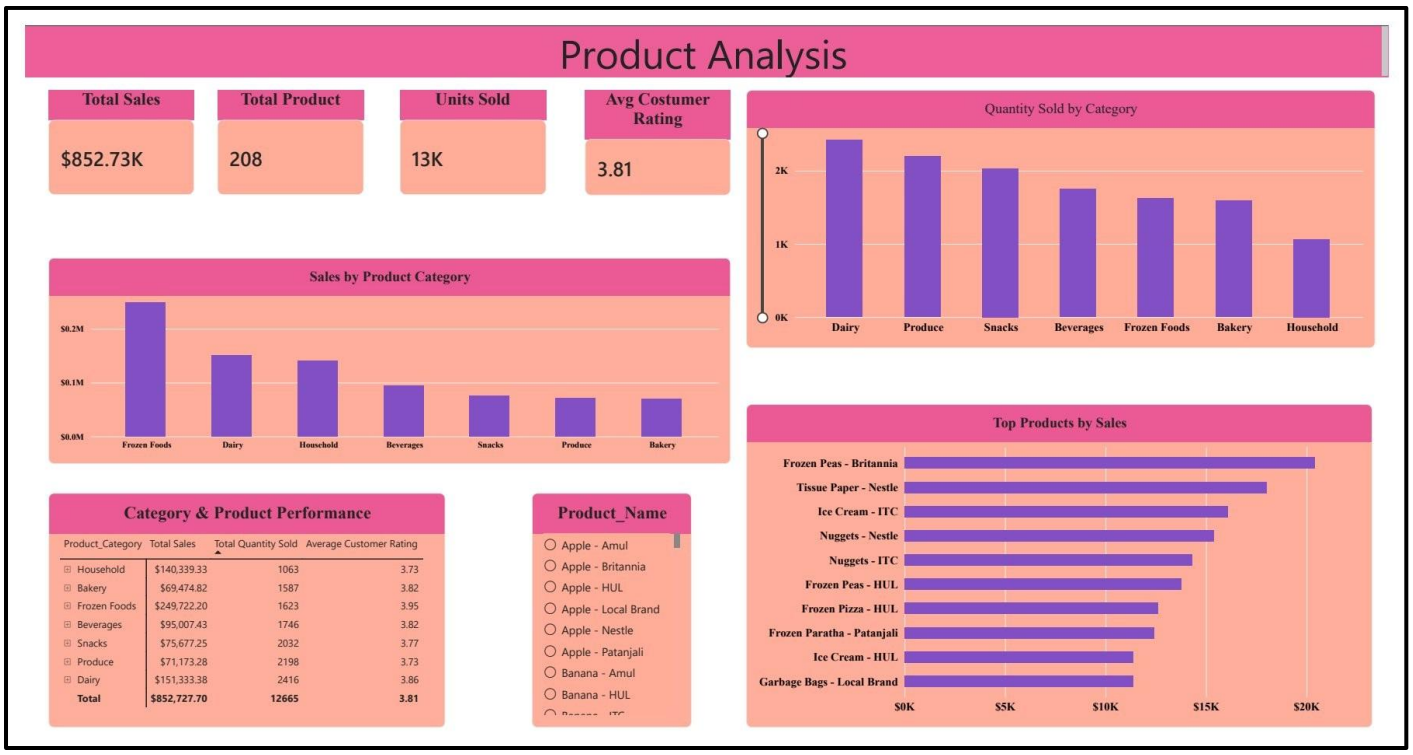


Figure 4.2: Product Analysis Dashboard in Power BI

Section 5: Advanced Power BI Analysis

5.1 Introduction

Advanced Power BI analysis helps examine grocery sales data beyond basic sales totals and product comparisons. In this project, Microsoft Power BI is used to analyze store and employee performance, discounts, returned transactions, customer ratings, payment methods, and future sales trends.

The dashboards combine Key Performance Indicators (KPIs), charts, tables, slicers, and advanced visuals to present different aspects of business performance. These visualizations help users identify patterns, compare results, and explore factors that may influence sales.

5.2 Objectives of Advanced Analysis

The main objectives of this section are:

- To evaluate sales performance across stores and employees.
- To analyze discounts and returned transactions.
- To examine transaction patterns and customer ratings.
- To identify factors associated with changes in sales.
- To explore historical sales trends and forecast future sales.
- To present detailed business information through interactive dashboards.

5.3 Dashboards Covered

This section presents the following advanced dashboards:

1. **Store and Employee Analysis:** Compares store sales and employee performance.
2. **Sales and Discount Analysis:** Examines discounts, returned transactions, and factors influencing sales.
3. **Customer and Transaction Analysis:** Studies payment modes, customer ratings, and transaction values.
4. **Future Sales Predictions:** Presents historical sales trends and forecast results for future periods.

These dashboards provide a broader view of the grocery business and complement the overall sales and product analysis presented in Section 4.

5.4 Store and Employee Analysis Dashboard

The Store and Employee Analysis Dashboard examines sales performance across different stores and employees. It presents key business indicators and visual comparisons to help identify differences in sales contribution.

Key components of the dashboard:

- **Total Stores:** Displays the number of stores included in the analysis.
- **Total Sales:** Shows overall sales of approximately $852.73K.
- **Total Employees:** Displays 99 employees.
- **Average Transaction Value:** Shows the average value of a transaction.
- **Sales by Store:** Compares sales across the five stores, S1 to S5.
- **Sales by Employee:** Ranks employees according to their sales contribution.
- **Sales Decomposition Analysis:** Breaks down total sales using dimensions such as store, product category, payment mode, employee name, and product name.

The Sales Decomposition Analysis visual allows users to explore how sales are distributed across multiple dimensions. This dashboard supports the comparison of store performance and employee contributions to overall sales.



Figure 5.1: Store and Employee Analysis Dashboard in Power BI

5.5 Sales and Discount Analysis Dashboard

The Sales and Discount Analysis Dashboard examines the relationship between sales performance, discounts, and returned transactions. It presents key indicators and visualizations that help users understand discount activity and compare return patterns across product categories.

Key components of the dashboard:

- **Total Sales:** Displays overall sales of approximately $852.73K.
- **Total Discount:** Shows a total discount value of approximately 56.27K.
- **Discounted Transactions:** Displays 288 discounted transactions.
- **Return Rate:** Shows a return rate of 7.80%.
- **Returned Transactions by Category:** Compares returned transactions across product categories.
- **Key Influencers of Total Sales:** Highlights factors associated with increases or decreases in sales.
- **Average Discount by Category:** Compares average discount values across categories.
- **Sales vs Average Discount Over Time:** Presents sales and average discount trends over time.

The Key Influencers visual helps explore factors associated with changes in total sales. The remaining visuals support comparisons of discounts and returns across categories and time periods.



Figure 5.2: Sales and Discount Analysis Dashboard in Power BI

5.6 Customer and Transaction Analysis Dashboard

The Customer and Transaction Analysis Dashboard examines transaction activity, payment methods, customer ratings, and transaction values. It presents key indicators and interactive visuals that help users understand customer-related patterns in the grocery sales dataset.

Key components of the dashboard:

- **Total Transactions:** Displays 500 transactions.
- **Average Transaction Value:** Shows an average transaction value of approximately 1.71K.
- **Average Customer Rating:** Displays an average rating of 3.81.
- **Returned Transactions:** Shows 39 returned transactions.
- **Transactions by Payment Mode:** Compares transaction counts across cash, credit card, debit card, UPI, and wallet payments.
- **Customer Rating by Product Category:** Compares average customer ratings across product categories.
- **Transaction Value vs Customer Rating:** Uses a scatter chart to explore the relationship between transaction value and customer ratings.
- **Interactive Filters:** Allow analysis by payment mode and year.

These visuals help users examine payment preferences, compare customer ratings, and explore how transaction values vary with customer ratings.

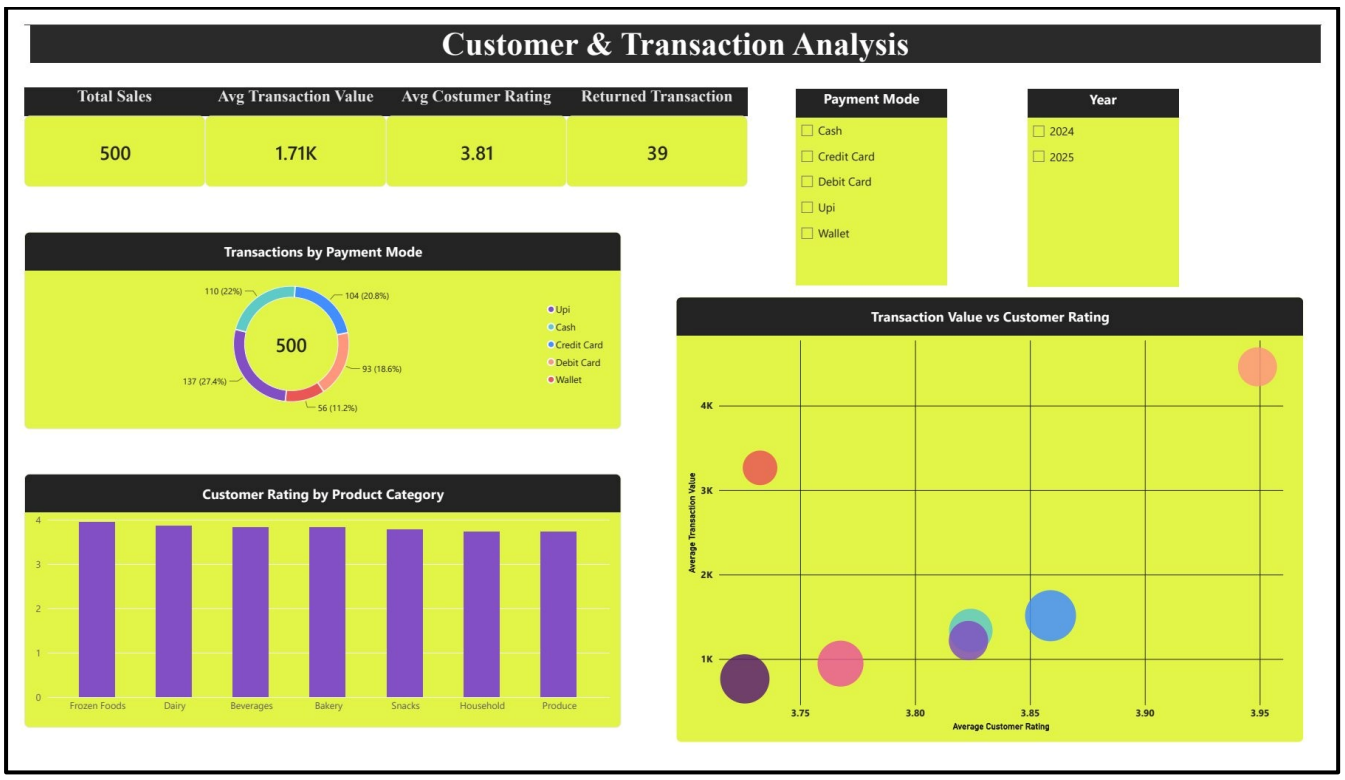


Figure 5.3: Customer and Transaction Analysis Dashboard in Power BI

5.7 Future Sales Predictions Dashboard

The Future Sales Predictions Dashboard uses historical grocery sales data to visualize sales trends and estimate future sales. Forecasting helps businesses understand possible future patterns and supports planning for inventory, product availability, and sales targets.

Key components of the dashboard:

- **Historical Sales Trend:** Displays changes in sales over the recorded period.
- **Sales Forecast:** Extends the historical sales trend to estimate future sales values.
- **Forecast Confidence Interval:** Where displayed, indicates a range of uncertainty around the forecast.
- **Time-Based Analysis:** Helps users observe changes in sales across different dates or periods.
- **Interactive Visualizations:** Allow users to explore the historical data and forecast results presented in the report.

The forecast provides an estimate based on historical patterns rather than a guarantee of future performance. Actual sales may differ because of changes in customer demand, discounts, product availability, and other business factors.

5.8 Importance of Sales Forecasting

Sales forecasting can support business planning by helping managers anticipate possible changes in demand, prepare inventory, and establish sales targets. When combined with historical sales analysis, it provides additional information for decision-making.

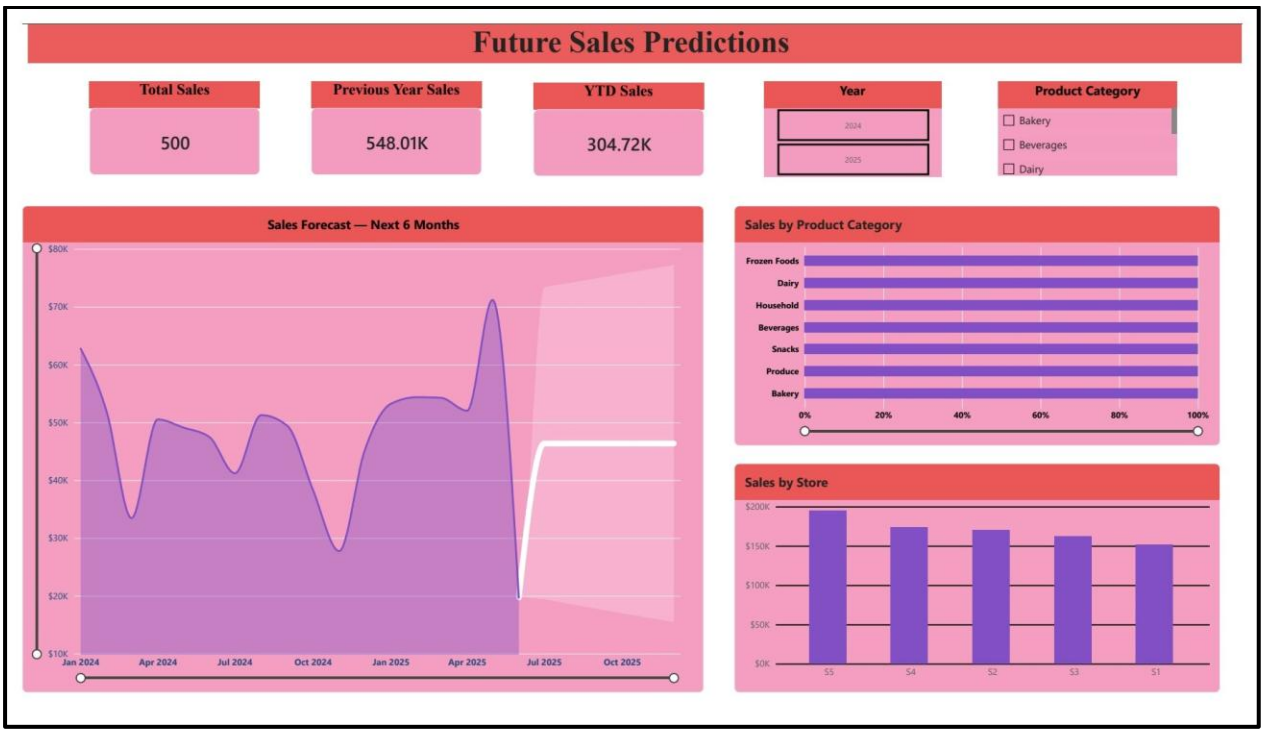


Figure 5.4: Future Sales Predictions Dashboard in Power BI

Section 6: Expected Outcomes, Learning Outcomes and Skills Developed

6.1 Expected Outcomes

The Grocery Sales Data Analysis project demonstrates how Microsoft Power BI can transform raw grocery sales data into meaningful business information. The expected outcomes of the project include:

- **Sales Performance Analysis:** Summarizing total sales, transactions, and quantities sold through KPIs.
- **Product and Store Comparison:** Identifying differences in sales performance across product categories and stores.
- **Discount and Return Analysis:** Examining discount patterns and returned transactions.
- **Customer and Transaction Insights:** Analyzing payment methods, transaction values, and customer ratings.
- **Sales Forecasting:** Exploring historical sales trends and estimated future sales.
- **Interactive Reporting:** Presenting information through dashboards, charts, and filters.

6.2 Learning Outcomes

After completing this project, the student is able to:

- Import, inspect, and prepare data using Power Query.
- Create relationships between tables in Power BI.
- Develop DAX measures for sales calculations and performance indicators.
- Design interactive dashboards to analyze business data.

6.3 Skills Developed

This project helps develop practical skills in data preparation, data modelling, analytical calculations, and dashboard development. It also improves the ability to interpret visualizations, compare business performance, and communicate findings through data-driven reports.

The project provides hands-on experience with Microsoft Power BI and demonstrates how Business Intelligence techniques can support business analysis and decision-making.

Section 7: Conclusion

The Grocery Sales Data Analysis project demonstrates the use of Microsoft Power BI to transform grocery sales data into meaningful information through data preparation, modelling, DAX calculations, interactive dashboards, and forecasting.

The project involved organizing the dataset, preparing data using Power Query, establishing relationships between tables, and creating DAX measures such as Total Sales, Year-to-Date (YTD) Sales, and Year-over-Year (YoY) Growth. These measures helped summarize sales performance and support the analysis of historical trends.

The dashboards developed for this project cover executive sales performance, product analysis, store and employee performance, discounts and returns, customer and transaction analysis, and future sales predictions. Different visualizations, KPI cards, and interactive filters make it easier to explore the data from multiple business perspectives.

Through this project, practical knowledge of data cleaning, data modelling, DAX, visualization, and sales forecasting was developed. The project also demonstrates how Business Intelligence tools can support performance monitoring, identify patterns, and assist business decision-making.

In conclusion, Microsoft Power BI provides an effective platform for analyzing grocery sales data and presenting the results in an organized and interactive format. The insights obtained from the report can support business planning and improve understanding of sales performance, while forecast results should be treated as estimates rather than guaranteed outcomes.

FOR FACULTY USE ONLY:

Correction Parameter	Formative Assessment [40%]	Timely Completion of practical [40%]	Attendance\ Learning Attitude [20%]	
Marks Obtained				